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4CS-B

1) A car travels 120 km in 2hrs and 180 km in 3hrs estimate the distance traveled after 2.5 hrs

2.) A store sells 500 items when the price is 100 and 350 items whe the price is 150. Estimate the number of item sold when the price is 125

3) A household consumes 200 litters of water at 5 persons, and 320 litters at 8 persons, estimate t consumption for 6 person

4) A machine consumes 40 kwh after operating for 2hr and 70kwh after 5hs. Estimate its consumption fo 4hs

CSP 110 - Week 9 - Seatwork

1. Car distance

$x_1 = 2$, $y_1 = 120$ $\quad y = y_1 + \dfrac{(x - x_1)(y_2 - y_1)}{x_2 - x_1}$

$x_2 = 3$, $y_2 = 180$

$x = 2.5$

$y = 120 + \dfrac{(2.5 - 2)(180 - 120)}{3 - 2}$

$= 120 + \dfrac{30}{1} = 150$

$\boxed{= \cancel{150}\ 150}$

2) Store

$x_1 = 100$, $y_1 = 500$

$x_2 = \cancel{350}\ 150$, $y_2 = 350$

$x = 125$

$y = \cancel{125}\ 500 + \dfrac{(125 - 100)(350 - 500)}{150 - 100}$

$y = 500 + (-75) = 425$

$\boxed{= 425}$

3) $x_1 = 5$, $y_1 = 200$

$\cancel{y}x_2 = 8$, $y_2 = 320$

$x = 6$

$$y = 200 + \frac{(6-5)(320-200)}{\cancel{x_2 - x_1}\ 8-5}$$

$y = 200 + 40$

$\boxed{y = 240}$

4) $x_1 = 2$, $y_1 = 40$

$x_2 = 5$, $y_2 = 70$

$x = 4$

$$y = 40 + \frac{(4-2)(70-40)}{5-2}$$

$= 40 + 20$

$\boxed{y = 60}$